

Software Requirements Specification (SRS)

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Project WikiMove – Migrating a Company Knowledge Base to a New Platform

IEEE Std 830-1998 Compliant Specification

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional, non-functional, and interface requirements for the WikiMove System in accordance with IEEE Std 830-1998 guidelines. The system serves as a project management, inventory tracking, triage classification, and governance platform to migrate an enterprise knowledge base of 14,000 legacy pages to a new modern knowledge platform before legacy licence expiration in 9 months (~39 weeks).

1.2 Scope

WikiMove encompasses the specification of workflows for:

- Ingesting and partitioning 14,000 legacy wiki pages across 12 engineering teams.
- Conducting tri-state triage (Keep, Archive, Delete) at a benchmark rate of 40 pages per person-day.
- Planning and tracking the migration of 7,000 retained pages (50% kept assumption) at 25 pages per person-day.
- Enforcing mandatory human content ownership (Zero Orphan Policy).
- Validating post-migration link integrity and search discoverability.
- Tracking mathematical schedule burn-down (630 total person-days, 15 person-days/week capacity, 42-week duration vs. 39-week licence limit).
- Managing long-term 90-day periodic content governance.

Academic Notice: Consistent with university instructions, WikiMove is an academic software engineering proposed system. Functional implementation is represented through static UI mockups; no live production database, backend, or cloud services are deployed.

1.3 Definitions, Acronyms, and Abbreviations

- BRD: Business Requirements Document
- BVA: Boundary Value Analysis
- CTO: Chief Technology Officer

- DRE: Defect Removal Efficiency
- FR: Functional Requirement
- IEEE: Institute of Electrical and Electronics Engineers
- MoSCoW: Must have, Should have, Could have, Won't have
- NFR: Non-Functional Requirement
- Person-Day (p-d): Unit of effort equal to one standard working day (8 hours) of one engineer.
- RMMM: Risk Mitigation, Monitoring, and Management Plan
- RTM: Requirements Traceability Matrix
- SRS: Software Requirements Specification
- Triage: Systematic process of inspecting content and categorizing into Keep, Archive, or Delete.
- TW: Technical Writer (Project Lead)

2. Overall Description

2.1 Product Perspective

WikiMove operates as an administrative management and governance layer positioned between the aging Legacy Wiki Platform and the Target Knowledge Platform.

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graph TD
    LegacyWiki["Legacy Wiki (14,000 Pages, Expires in ~39 Wks)"] --> WikiMove["WikiMove Management System (Proposed Architecture)"]
    WikiMove --> TargetPlatform["New Knowledge Platform (7,000 Kept Pages, Governed)"]
    WikiMove --> ArchiveStorage["Cold Storage (Archived Compliance Stubs)"]
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2.2 User Classes & Characteristics

1. Technical Writer (Project Lead): Power user; coordinates overall migration, oversees orphan reassignments, audits QA logs, monitors schedule gap. (Allocation: 3 person-days/week).
2. Engineering Team Members (12 Teams): Domain reviewers; triage subsystem pages, update content, assign named owners. (Allocation: 1 person-day/week per team = 12 person-days/week).
3. Content Owners: Named custodians; accept accountability for specific kept pages and execute 90-day periodic revalidations.
4. Project Manager: Monitors WBS, milestones, critical path, risk exposures, and resource constraints.
5. QA / Test Engineer: Validates link integrity, verifies search indexing, logs defect density.
6. CTO / Management: Executive stakeholder; reviews dashboard KPIs and enforces organizational compliance.

2.3 General Constraints & Assumptions

- Constraint C-01: The project must achieve decommission readiness within 9 months (approximately 39 weeks).
- Constraint C-02: Weekly effort cannot exceed 15 person-days per week (3 TW + 12 Engineering).
- Constraint C-03 (Academic): System must not require live production backend, database, or API deployment.
- Assumption A-01: Exactly 50% of the 14,000 pages (7,000 pages) will be designated as Keep.
- Assumption A-02: Triage review velocity is fixed at 40 pages per person-day.
- Assumption A-03: Migration velocity is fixed at 25 pages per person-day.

3. External Interface Requirements

3.1 User Interfaces

The system provides 6 responsive, accessible, card-based static screens adhering to light-theme aesthetics:

- Screen 1 (Dashboard): High-level metrics, progress bars, licence timer, top risk card.
- Screen 2 (Page Triage): Filterable table of assigned pages with Keep, Archive, and Delete controls.
- Screen 3 (Page Review): Split-view content inspector, metadata display, reviewer notes, and owner binding.
- Screen 4 (Migration Status): Batch tracking for the 7,000 kept pages, validation status, error log.
- Screen 5 (Ownership Management): Content custody registry displaying assigned owners and 90-day review dates.
- Screen 6 (Risks & Schedule): Quantitative risk matrix (R1-R4) and 42-week vs. 39-week schedule breakdown.

3.2 Hardware & Communication Interfaces

- Standard client web browsers (Chrome, Safari, Firefox, Edge) supporting HTML5, CSS3, and modern ECMAScript.
 - Standard HTTP/HTTPS network communication (zero specialized hardware required).
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4. Functional Requirements

FR-01: Page Inventory Management

- Description: The system shall ingest and index metadata for all 14,000 legacy wiki pages.
- Inputs: Legacy platform page export (Page ID, Title, URL, Creation Date, Last Modified, Subsystem Tag).
- Processing: Parse metadata, assign unique WikiMove ID (WP-xxxxx), flag age (> 3 years = potentially outdated).
- Outputs: Searchable centralized inventory viewable on Dashboard and Triage screens.

FR-02: Page Review Interface

- Description: The system shall display legacy page content, active hyperlinks, revision history, and author metadata for inspection.
- Inputs: Page ID selection from Triage queue.
- Processing: Render HTML/Markdown content preview, extract internal and external link targets.
- Outputs: Rendered document display on Screen 3.

FR-03: Page Classification Engine

- Description: The system shall enable reviewers to assign exactly one terminal triage classification to each page: Keep, Archive, or Delete.
- Inputs: Reviewer action button click (Keep, Archive, Delete) and mandatory justification comment.
- Processing: Validate reviewer authorization; verify that daily triage throughput tracks against the 40 pages/day benchmark.
- Outputs: State transition recorded; page status updated in inventory.

FR-04: Keep Page Workflow

- Description: The system shall transition valid, accurate pages to Kept status, qualifying them for migration.
- Inputs: Reviewer selects Keep classification.
- Processing: Verify content is current; enforce mandatory input of a valid Content Owner. Total kept volume capped at 50% baseline (7,000 pages).
- Outputs: Page queued in KeptPendingMigration state.

FR-05: Archive Page Workflow

- Description: The system shall transition historical or compliance-mandatory pages to Archived status.
- Inputs: Reviewer selects Archive with retention rationale (e.g., Legal, Audit, Deprecated Architecture).
- Processing: Generate read-only archival snapshot; create a redirection stub for legacy URLs.
- Outputs: Page marked Archived; removed from migration queue.

FR-06: Delete Page Workflow

- Description: The system shall flag obsolete, redundant, or empty pages for permanent decommissioning.
- Inputs: Reviewer selects Delete.
- Processing: Apply a 14-day soft-delete grace window; generate dead-link redirection notices.
- Outputs: Page marked MarkedForDeletion; permanently purged following grace expiration.

FR-07: Content Owner Designation (Zero Orphan Policy)

- Description: The system shall prohibit any page from migrating unless an active, named employee is assigned as Content Owner.
- Inputs: Content Owner employee ID, email, and engineering squad.
- Processing: Validate employee active status against organizational directory. Flag unassigned pages as Orphan Alert.
- Outputs: Owner bound to page metadata; automated welcome notification sent to owner.

FR-08: Migration Batch Planning

- Description: The system shall group the 7,000 kept pages into sprint migration batches sized according to team capacity.

- Inputs: Domain category, target batch size, scheduled sprint window.
- Processing: Calculate required effort using benchmark migration velocity of 25 pages per person-day (\$7,000 / 25 = 280\text{ person-days}\$).
- Outputs: Scheduled migration batches displayed on Screen 4.

FR-09: Migration Progress Tracking

- Description: The system shall track the live execution status of every migration batch across states: Queued, In Progress, Migrated, Validation Pending, and Certified.
- Inputs: Batch execution updates.
- Processing: Compute real-time completion %, velocity variance, and remaining person-days.
- Outputs: Dynamic progress bars and burn-down metrics.

FR-10: Link Integrity Validation

- Description: The system shall validate all hyperlinks within migrated pages to prevent broken links post-cutover.
- Inputs: Migrated page content.
- Processing: Scan anchor tags; match legacy intra-wiki URLs against the WikiMove rewrite registry. Verify HTTP 200 response on external targets.
- Outputs: Link validation status (Certified vs. Broken Link Defect Logged).

FR-11: Search Discoverability Validation

- Description: The system shall verify that every migrated page is successfully indexed and discoverable in the target platform's search engine.
- Inputs: Page title, keywords, and domain tags.
- Processing: Execute synthetic test queries against target search index; verify page surfaces within top 5 search results.
- Outputs: Search index certification status.

FR-12: Project Progress & Schedule Tracking

- Description: The system shall monitor cumulative project effort and compare completion forecasting against the 39-week licence expiration deadline.
- Inputs: Cumulative triage person-days and migration person-days expended.
- Processing: Compute: Total Effort = $\$350 + 280 = 630\text{ p-d}$. Duration = $\$630 / 15 = 42\text{ weeks}$. Schedule Gap = $\$42 - 39 = +3.0\text{ weeks}$.
- Outputs: Visual schedule gap alerts on Dashboard (Screen 1) and Risks screen (Screen 6).

FR-13: Risk Tracking & RMMM Integration

- Description: The system shall maintain an active risk log for all identified project risks, calculating Exposure ($\$E = P \times I$) and maintaining mitigation triggers.
- Inputs: Risk IDs (R1, R2, R3, R4), Probability, Impact.
- Processing: Calculate:
 - $\$R2\text{ (Teams no review)} = 0.6 \times 7 = 4.2\text{ (Rank 1)}$
 - $\$R1\text{ (Licence expires)} = 0.4 \times 9 = 3.6\text{ (Rank 2)}$
 - $\$R3\text{ (Broken links)} = 0.4 \times 4 = 1.6\text{ (Rank 3)}$
 - $\$R4\text{ (No owner)} = 0.5 \times 1 = 0.5\text{ (Rank 4)}$
- Outputs: Risk exposure rankings, mitigation owner assignments, and contingency status.

FR-14: Management Reporting & KPI Analytics

- Description: The system shall generate consolidated progress summaries for the CTO, Project Manager, and Technical Writer.
- Inputs: Aggregated triage, migration, QA, and ownership logs.
- Processing: Compile completion percentages, defect removal efficiency (DRE), defect density, and team quota adherence.
- Outputs: Printable executive summaries and real-time dashboard visualizations.

FR-15: Post-Migration Ownership & Governance

- Description: The system shall enforce a sustainable post-migration content lifecycle by executing automated 90-day revalidation notifications.
- Inputs: Last reviewed timestamp.
- Processing: Check if current date exceeds last review date + 90 days. If true, flag page as Review Due and

send owner prompt.

- Outputs: Governance alert queue displayed on Screen 5.

5. Non-Functional Requirements (NFRs)

NFR ID	Requirement Name	Description	Measurement / Metric	Target Value *(Project Assumption)*	Verification Method
NFR-01	Link Integrity	All internal intra-wiki hyperlinks within migrated pages must resolve correctly without 404 dead links.	Broken link ratio: (Broken Links / Total Links) × 100	0.0% Broken Internal Links *(Target)*	Automated automated link crawler testing on all 7,000 kept pages.
NFR-02	Search Performance	Search queries on the target platform must return relevant indexed pages with sub-second response time.	Search query response latency (95th percentile)	< 1.0 second response time *(Assumption)*	Synthetic keyword search query latency profiling across 100 test terms.
NFR-03	Availability	The WikiMove management portal interface shall remain accessible during business operating hours.	System uptime during business hours (08:00 - 18:00 UTC)	99.5% Uptime *(Assumption)*	Web server ping monitoring and availability logs.
NFR-04	Usability	The triage and review interface must allow engineers to execute triage decisions rapidly without cognitive overload.	Mean time to complete a single page review & triage decision	< 12 minutes per page (enabling 40 pages/day quota)	Usability trial review sessions with sample engineering team members.
NFR-05	Reliability	The system shall prevent data corruption or lost review decisions during network interruptions.	Decision submission failure rate	< 0.1% failure rate *(Assumption)*	Client-side form auto-save and transactional submission verification.
NFR-06	Security	Access to triage classification and owner reassignment must be restricted based on authenticated employee roles.	Role-Based Access Control (RBAC) validation	100% unauthorized action rejection	Role permission penetration testing and audit trail verification.
NFR-07	Maintainability	System UI components and document schemas must be modular, adhering to clean HTML5/CSS standards.	W3C validation compliance and code modularity	100% W3C valid markup	Static code analysis and HTML/CSS linters.
NFR-08	Auditability	Every classification, owner assignment, and status transition must be logged with an immutable audit trail.	Audit log completeness ratio	100% captured audit events	Audit record inspection against test scenario action sequences.
NFR-09	Data Integrity	Content formatting, code snippets, and table structures must be preserved without loss during migration.	Formatting fidelity comparison score	> 99.0% visual & semantic parity	Side-by-side DOM diff testing between legacy and target page renders.

NFR-10	Traceability	Every functional requirement must trace directly to business needs, use cases, and verification test cases.	RTM mapping coverage percentage	100% bidirectional traceability	Inspection of Requirements Traceability Matrix (RTM).
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6. Requirements Traceability Matrix (RTM)

Req ID	Requirement Description	Type	Priority	Business Need	Use Case	Test Case ID	Acceptance Criteria
FR-01	Wiki Page Inventory & Ingestion	Functional	Must	BO-01	UC-01	TC-01	Ingests 14,000 legacy records with accurate metadata.
FR-02	Page Review Interface	Functional	Must	BO-01	UC-02	TC-02	Displays content preview, links, and metadata cleanly.
FR-03	Page Classification Engine	Functional	Must	BO-01	UC-03	TC-03	Records Keep, Archive, or Delete with reviewer comment.
FR-04	Keep Page Workflow	Functional	Must	BO-02	UC-03	TC-04	Transitions page to Kept; caps at 50% baseline (7,000 pg).
FR-05	Archive Page Workflow	Functional	Should	BO-01	UC-03	TC-05	Generates archival stub; removes page from active migration.
FR-06	Delete Page Workflow	Functional	Should	BO-01	UC-03	TC-06	Enforces 14-day grace window before permanent purge.
FR-07	Assign Content Owner	Functional	Must	BO-03	UC-04	TC-07	Prohibits unassigned pages from migrating (Zero Orphan).
FR-08	Migration Batch Planning	Functional	Must	BO-02	UC-05	TC-08	Batches 7,000 kept pages at 25 pages/ person-day velocity.
FR-09	Migration Progress Tracking	Functional	Must	BO-02	UC-06	TC-09	Tracks progress percentage and batch state transitions.
FR-10	Link Integrity Validation	Functional	Must	BO-04	UC-07	TC-10	Confirms zero broken internal hyperlinks (NFR-01).

FR-11	Search Validation	Functional	Should	BO-04	UC-08	TC-11	Verifies migrated pages index and return in search queries.
FR-12	Project Progress Tracking	Functional	Must	BO-05	UC-10	TC-12	Flags 3-week gap (42-week baseline vs 39-week licence).
FR-13	Risk Tracking & RMMM	Functional	Must	BO-05	UC-09	TC-13	Calculates exposure ($\$E = P \times \I); ranks R2 (4.2) > R1 (3.6).
FR-14	Reporting & KPI Analytics	Functional	Should	BO-01	UC-11	TC-14	Generates executive summaries for CTO and Technical Writer.
FR-15	Post-Migration Governance	Functional	Must	BO-06	UC-11	TC-15	Enforces 90-day periodic revalidation notifications.
NFR-01	Link Integrity Metric	NFR	Must	BO-04	UC-07	TC-10	0.0% broken intra-wiki links post-migration.
NFR-02	Search Performance Metric	NFR	Should	BO-04	UC-08	TC-11	Query response latency < 1.0 second on target platform.
NFR-04	Usability Throughput	NFR	Must	BO-01	UC-02	TC-02	Supports 40 pages/person-day triage benchmark.